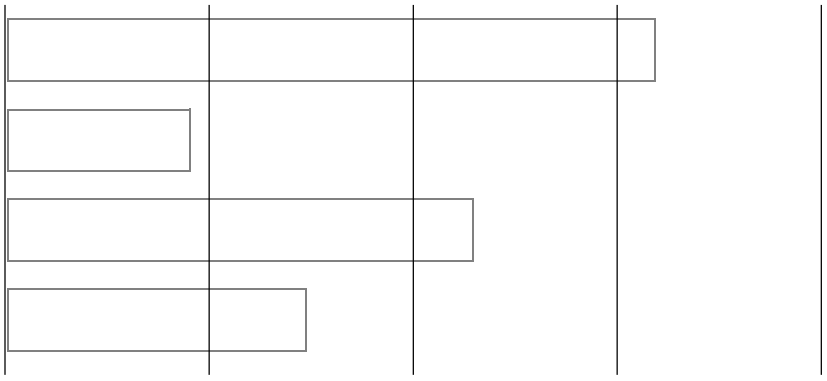
MENU

Industrial instrument technicians and mechanics (NOC 2243)

Minimum education: Diploma, Certificate or Apprenticeship Training

- Average salary
- Occupation size
- Job stability
- Demand growth



Profile last updated: March 23, 2017

01 Overview

Industrial instrument technicians and mechanics repair, maintain, calibrate, adjust, and install the dials, sensors and other instrumentation that measures and controls machinery in industrial and commercial plants. Instrumentation gives operators information about the condition and operation of machinery, ensuring the safety and function of the plant or system.

People in this occupation:

- are employed by pulp and paper processing companies, hydro power generating companies, mining, petrochemical and natural gas companies, industrial instrument and other manufacturing companies, health services and industrial instrument servicing establishments
- benefit from having a knowledge of electronics, physics, mathematics and chemistry
- should have an interest in the scientific and technical aspects of machinery and industrial processes
- must have troubleshooting abilities, problem solving skills and an attention to detail
- must have a well-rounded knowledge of industrial production processes, controls and power systems such as pneumatics, electro-pneumatics, hydraulics, electricity,

electronics, computers and networks

Common job titles

- mechanic, industrial / precision
- mechanic, utilities / maintenance
- repairer, industrial instrument panel
- repairer, industrial process control equip
- repairer, nucleonic controllers
- repairer, process control equipment
- technician, heavy water plant control
- technician, industrial instrumentation
- technician, nuclear generating station

02 Earnings

Provincial median salary

$80,727

Source: Full-time, full-year median employment income, 2011 National Household Survey

Provincial hourly rate

Low	N/A
Median	N/A
High	N/A

Source: 2016 Job Bank Wage Report

03 Duties

Industrial instrument technicians and mechanics:

- consult manufacturer's manuals, circuit diagrams and blueprints to determine tests and maintenance procedures for instruments used for measuring and controlling flow, level, pressure, temperature, chemical composition and other variables in manufacturing and processing
- inspect and test operation of instruments and systems to diagnose faults using pneumatic, electrical and electronic testing devices and precision measuring instruments
- repair and adjust system components, such as sensors, transmitters, final control elements, distributed computer control systems (DCS) and programmable logic controllers, or remove and replace defective parts
- calibrate components and instruments according to manufacturers' specifications
- perform scheduled preventive maintenance work and complete test and maintenance

reports
- install control and measurement instruments on existing and new plant equipment and processes
- consult with and advise process operators to troubleshoot process and system faults

04 Work environment

Industrial instrument technicians and mechanics conduct work on large production machines in industrial facilities such as refineries, mills or power plants.

These individuals may perform tasks in confined spaces, high spaces or areas exposed to noise, fumes or high heat levels. Workers may also be required to work outdoors with exposure to varied weather. Safety precautions minimize risks, which may include electric shock, exposure to dangerous chemicals or materials under high pressure.

Employment is generally full time and year round. These technicians and mechanics work regular hours or shift work and may be called upon during emergencies, such as when manufacturing and power control systems instrumentation fails to work. During emergencies, equipment may be damaged, fires have the potential to break out and valuable raw materials could be lost, so workers are often required to concentrate in urgent or stressful situations.

Workers must be familiar with developments in their area of specialty and often retrain or upgrade skills to keep up with rapid changes in technology.

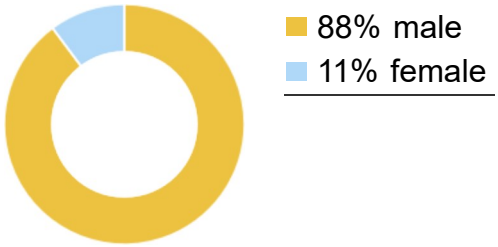
05 Workforce and employment statistics

Workforce characteristics

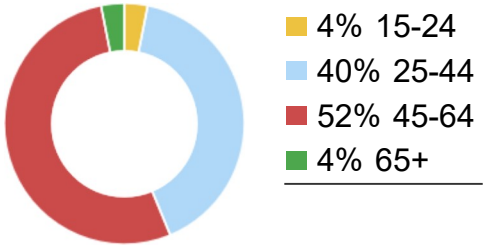
700 workers are employed

69 % of workers are working mostly full time

Employment by gender



Labour force by age group



Source: 2011 National Household Survey

Labour force by industry

Manufacturing 60%

Mining and Oil and Gas Extraction 11%

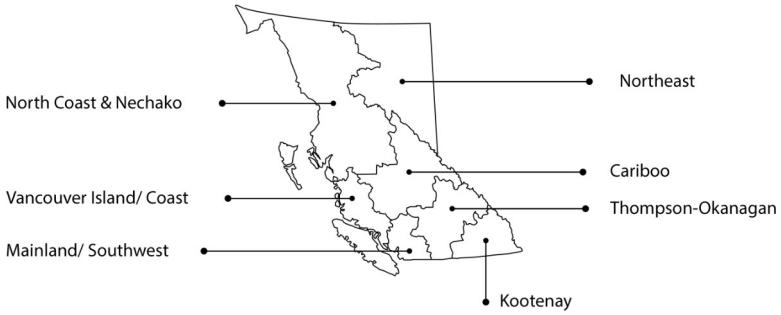
Professional, Scientific, and Technical Services 7%

Construction 7%

Wholesale and Retail Trade 5%

Source: 2011 National Household Survey

Employment by region



Vancouver Island / Coast

11.6%

16.1%

Mainland / Southwest

37.7%

62.8%

Thompson-Okanagan

2.9%

10.8%

Kootenay

14.5%

3.4%

Cariboo

13.8%

3.5%

North Coast & Nechako

2.9%

1.7%

Northeast

16.7%

1.7%

% employment of this occupation

% employment of all occupations

Source: 2011 National Household Survey

06 Job requirements

Education, training and qualifications

Industrial instrument technicians and mechanics must complete secondary school, including Math 11, and physics and chemistry courses are also recommended.

In B.C., there is no requirement that industrial instrument technicians or mechanics be certified, however, certification or apprenticeship can offer more well-rounded training and possibly increased employment opportunities.

The Industry Training Authority's Certificate of Qualification as an industrial instrument mechanic requires completion of a four-year program of in-school and work-based training or a four-year apprenticeship program in industrial instrumentation. Those who complete a formal apprenticeship also receive a Certificate of Apprenticeship.

The final assessment for each of these routes is the Inter-provincial Red Seal written examination.

Certified Journeypersons may apply to the Applied Science Technologists and Technicians of British Columbia (ASTTBC) for registration as Certified Technicians and may be required to take additional coursework (i.e. math, communications, business).

Individuals with at least seven years of full-time experience as an industrial instrument technician or mechanic or qualifications from another province may write a challenge exam. Those who pass this exam will receive the B.C. Certificate of Qualification and the Interprovincial Standard Red Seal Endorsement. Individuals working in the oil and gas industry should also have completed H2S Alive and Industrial First-Aid, Level 1

For more information please see the Industry Training Authority website at: www.itabc.ca.

Skills

- Numerical Ability
- Detail-Oriented

- Methodical
- Object-Oriented

- Verbal & Written Comprehension
- Manual Dexterity

07 Subject Areas & Training Resources

EducationPlannerBC

www.educationplannerbc.ca

Trades/Apprenticeship Resources

www.workbc.ca/trades

Subject Area

Engineering and Electronics

- Instrumentation Technology Related
- Computer Engineering Related

08 Career paths

Recent graduates typically obtain positions related to factory maintenance. Technicians and mechanics with experience and further education may progress to supervisory positions, provide consulting services or become trainers or educators. Industrial instrument mechanics can also apply for dual journey status in both instrumentation and electrical.

09 Employment outlook

N/A - Data not available or not provided due to data quality issues

Provincial Outlook:

Unemployment rate

2015	5.4%
2020	4.9%
2025	4.9%

Job openings

2015	40
2020	30
2025	10

-0.4% Expected annual demand growth rate (2015-2025)
220 Job openings (2015-2025)

N/A - Data not available or not provided due to data quality issues

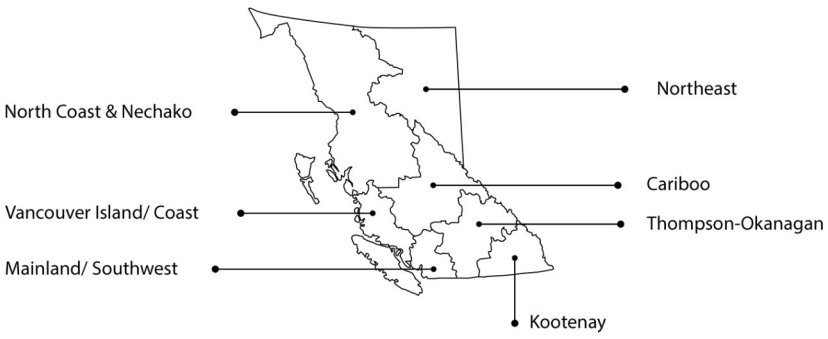
Occupation Outlook for B.C.:

% of job openings from replacement **100%**

% of job openings from expansion **0%**

Regional Outlook

N/A - Data not available or not provided due to data quality issues



View regional outlook for:

British Columbia

Expected number of job openings (2015-2025)	**220**
Average annual employment growth (2015-2025)	**-0.4%**
Expected increase in employment (2015-2025)	**N/A**

Cariboo

Expected number of job openings (2015-2025)	**30**
Average annual employment growth (2015-2025)	**-0.7%**
Expected increase in employment (2015-2025)	**-5**

Kootenay

Expected number of job openings (2015-2025)	**60**
Average annual employment growth (2015-2025)	**1.2%**
Expected increase in employment (2015-2025)	**15**

Mainland / Southwest

Expected number of job openings (2015-2025)	**60**
Average annual employment growth (2015-2025)	**0.0%**
Expected increase in employment (2015-2025)	**0**

North Coast & Nechako

Expected number of job openings (2015-2025)	**N/A**
Average annual employment growth (2015-2025)	**N/A**
Expected increase in employment (2015-2025)	**N/A**

Northeast

Expected number of job openings (2015-2025)	**90**
Average annual employment growth (2015-2025)	**1.6%**
Expected increase in employment (2015-2025)	**25**

Thompson-Okanagan

Expected number of job openings (2015-2025)	**N/A**
Average annual employment growth (2015-2025)	**N/A**
Expected increase in employment (2015-2025)	**N/A**

Vancouver Island / Coast

Expected number of job openings (2015-2025)	**N/A**
Average annual employment growth (2015-2025)	**N/A**
Expected increase in employment (2015-2025)	**N/A**

10 Insights from industry

Job openings will arise due to both retirement and the creation of new jobs. However, due to the relatively small number of individuals employed in this group, the total number of new openings that arise in the upcoming years will be relatively few.

There is expected to be increased demand in this area as a number of new projects are anticipated. There is presently a shortage of skilled workers and new graduates available in B.C. to fill openings that arise. However, companies in Alberta often offer to transport workers across the border and charge a lower rate. This helps to supply the demand, but may also take positions away from local workers.

The industrial instrumentation industry reports an urgent need for more skilled and qualified control technicians. There is currently a high employment rate for graduates of these programs.

Demand is reported to be in the pulp and paper, and mining, oil and gas industries. New jobs are also expected to also open up in instrument service and sales companies, the heating and ventilating field, and in environmental and pollution control.

The most successful individuals in this occupation will focus on professional development and keeping up-to-date with changes in technology.

11 Additional resources

- Industry Training Authority
  www.itabc.ca
- Applied Science Technologists & Technicians of British Columbia
  www.asttbc.org/
- Interprovincial Standards Red Seal Program
  www.red-seal.ca/
- Northern Opportunities: Industrial Instrument Mechanic description
  northernopportunities.bc.ca/the-programs/trades-programs/industrial-instrument-mechanic/
- International Society of Automation
  www.isa.org

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